CP violation and Unitarity triangle

$\cancel{CP} \longleftrightarrow$ CKM matrix

3 types of $\cancel{CP}$:

1) direct $\quad i \to f \neq \bar{i} \to \bar{f}$

2) in mixing: $\quad P^0 \to \bar{P}^0 \neq \bar{P}^0 \to P^0$

$P^0 = D^0, K^0, B^0, B_s^0$

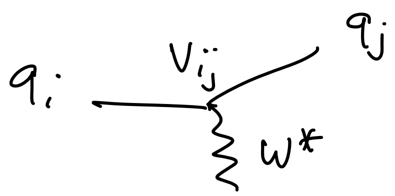
3) in interference

$A_1: \; P^0 \longrightarrow f$

$A_2: \; P^0 \to \bar{P}^0 \to f$

$A_{tot} = A_1 + A_2$ is sensitive to complex phase in matrix elements.

CKM matrix $\quad V_{ij}$ elements



$$V_{CKM} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$

$$V_{CKM}^{\dagger} V_{CKM} = \mathbb{1} \implies 9 \text{ conditions.} \qquad \mathbb{1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$V_{CKM}^{\dagger} = \begin{pmatrix} V_{ud}^* & V_{cd}^* & V_{td}^* \\ V_{us}^* & V_{cs}^* & V_{ts}^* \\ V_{ub}^* & V_{cb}^* & V_{tb}^* \end{pmatrix}$$

$$V_{ud} V_{ud}^* + V_{us} V_{us}^* + V_{ub} V_{ub}^* = 1$$

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$$

$$\left.\begin{array}{l} V_{ud} \simeq \cos\theta_c \\ V_{us} \simeq \sin\theta_c \end{array}\right\} \implies |V_{ub}|^2 \ll 1$$

$$|V_{cd}|^2 + |V_{cs}|^2 + |V_{cb}|^2 = 1$$

$$|V_{td}|^2 + |V_{ts}|^2 + |V_{tb}|^2 = 1$$

$$V_{\text{CKM}} = \begin{pmatrix} 1-\lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1-\lambda^2/2 & A\lambda^2 \\ A\lambda^3(1-\rho-i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

Wolfenstein parameterization

$$V_{ud}^* V_{us} + V_{cd}^* V_{cs} + V_{td}^* V_{ts} = 0 \qquad \lambda \quad \lambda \quad \lambda^5$$

$$\boxed{V_{ub}^* V_{ud} + V_{cb}^* V_{cd} + V_{tb}^* V_{td} = 0} \qquad \underline{\lambda^3 \quad \lambda^3 \quad \lambda^3}$$

$$V_{us}^* V_{ub} + V_{cs}^* V_{cb} + V_{ts}^* V_{tb} = 0 \qquad \lambda^4 \quad \lambda^2 \quad \lambda^2$$

$$V_{ud}^* V_{td} + V_{us}^* V_{ts} + V_{ub}^* V_{tb} = 0 \qquad \lambda^3 \quad \lambda^3 \quad \lambda^3$$

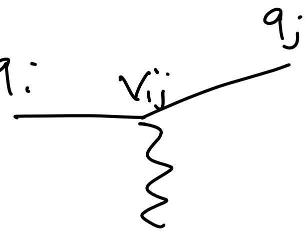
$$V_{td}^* V_{cd} + V_{ts}^* V_{cs} + V_{tb}^* V_{cb} = 0 \qquad \lambda^4 \quad \lambda^2 \quad \lambda^2$$

$$V_{ud}^* V_{cd} + V_{us}^* V_{cs} + V_{ub}^* V_{cb} = 0 \qquad \lambda \quad \lambda \quad \lambda^5$$

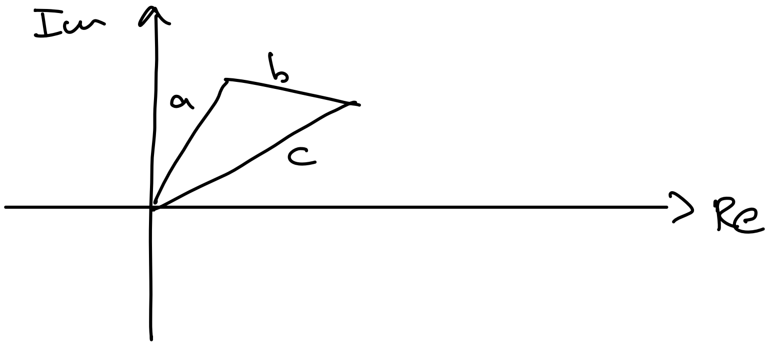
$V_{ij}^* V_{ik}$: Complex

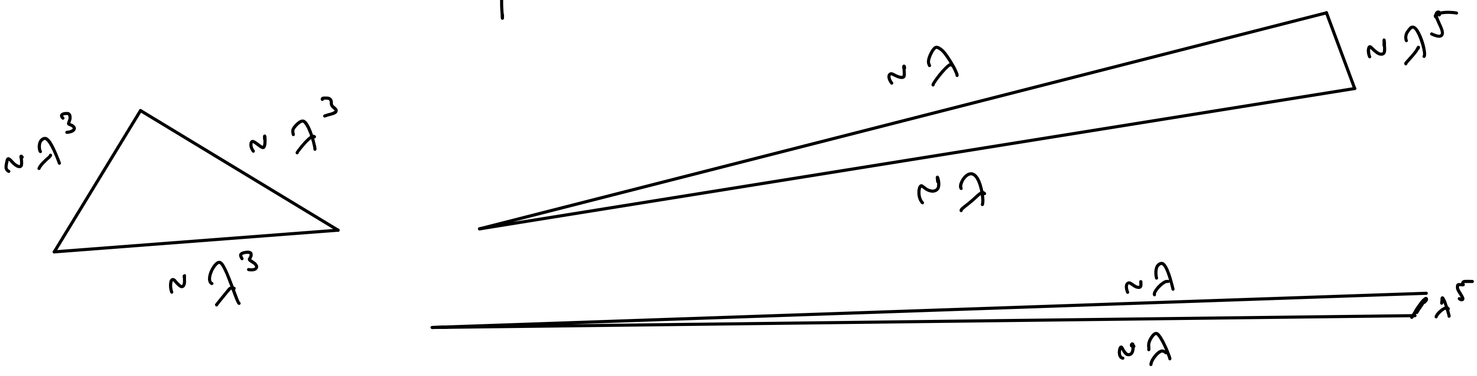
$\lambda \simeq \sin\theta_c$

V_{ij}



$a + b + c = 0$ in Complex plane



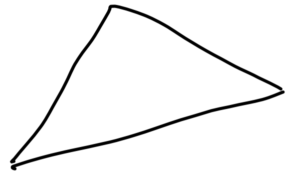


Relations with terms of same order of λ are easier to measure experimentally.

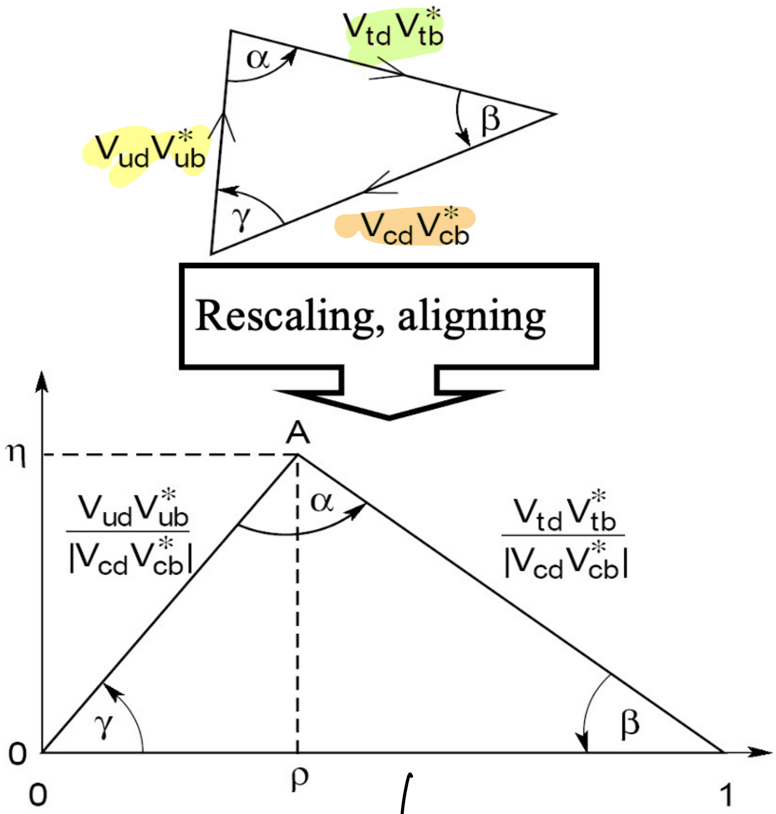
The line with V_{xb} can be accessed by studying processes with B mesons.

$\Rightarrow$ need to produce millions of B mesons.

$$V_{ub}^* V_{ud} + V_{cb}^* V_{cd} + V_{tb}^* V_{td} = 0$$



$V_{ud} \simeq \cos\theta_c$ $\qquad$ $V_{tb} \simeq 1$



$$\frac{V_{cd}V_{cb}^*}{|V_{cd}V_{cb}^*|}$$

Angles of Unitarity Triangle

$$\alpha = \phi_2 \equiv \arg\left[-\frac{V_{td}V_{tb}^*}{V_{ud}V_{ub}^*}\right],$$

$$\beta = \phi_1 \equiv \arg\left[-\frac{V_{cd}V_{cb}^*}{V_{td}V_{tb}^*}\right],$$

$$\gamma = \phi_3 \equiv \arg\left[-\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*}\right]$$

$$\begin{array}{cc} & V_{ub} \\ \begin{pmatrix} 1 & 1 & e^{-i\gamma} \\ 1 & 1 & 1 \\ e^{-i\beta} & 1 & 1 \end{pmatrix} & \\ V_{td} & \end{array}$$

$\alpha + \beta + \gamma = \pi$

If experimentally measure $\alpha, \beta, \gamma \neq 0$.

$\Rightarrow$ triangle exists.

$\Rightarrow$ CKM unitarity holds

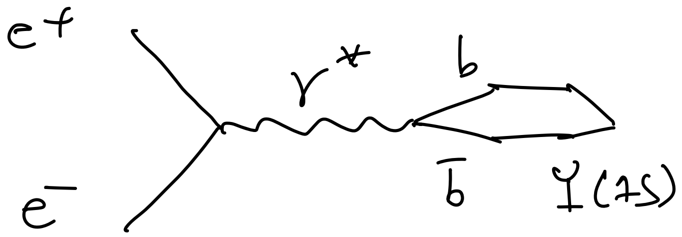
$\Rightarrow$ CKM has 3 real param. + 1 complex phase.

Complex phase necessary to explain $C\!\!\!/P$ in weak interactions

$\Rightarrow$ study CP violation in B mesons.

$\Rightarrow$ Need a B-Factory.

CP Violation in B mesons



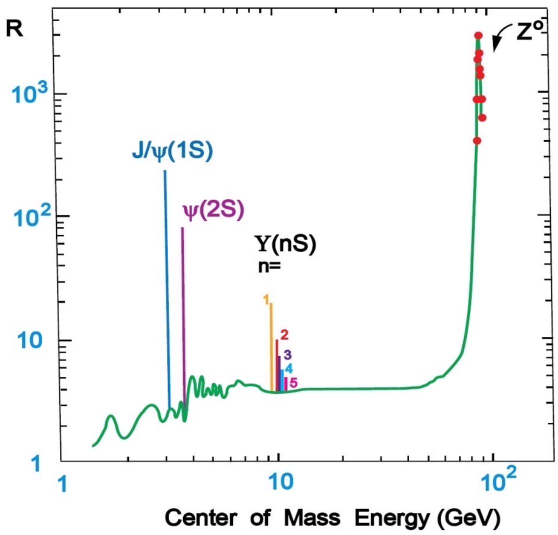
$\sqrt{s} \geq 2m_b$

$e^+e^- \to \Upsilon(1S) \to \mu^+\mu^-$
e^+e^-

$\sqrt{s} = m_{\Upsilon(1S)} \gtrsim 9$ GeV

$J^{PC} = 1^{--}$

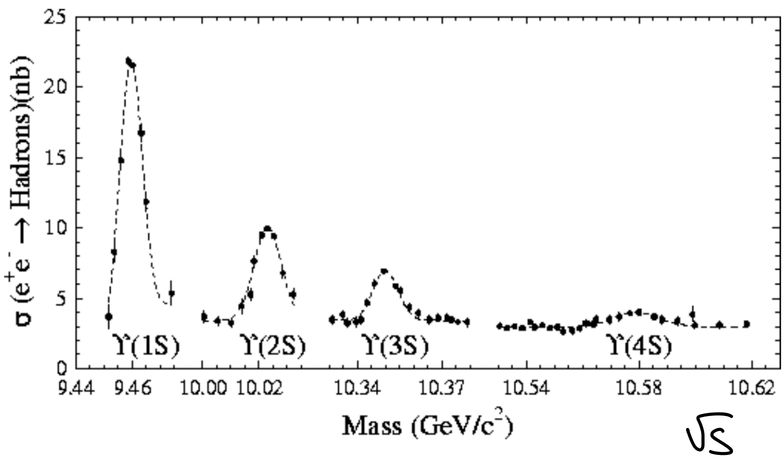
$$R = \frac{\sigma(e^+e^- \to had)}{\sigma(e^+e^- \to \mu^+\mu^-)}$$



$e^+e^- \to \Upsilon(1S) \to$

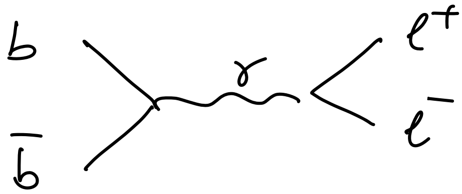
$m_B = 5.279$ GeV

$m_{\Upsilon(1S)} = 9.46$ GeV

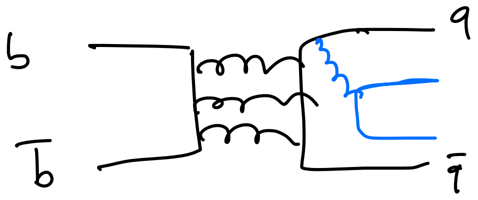


$\sqrt{s}$

$\Upsilon(1S) \not\to$ B mesons.



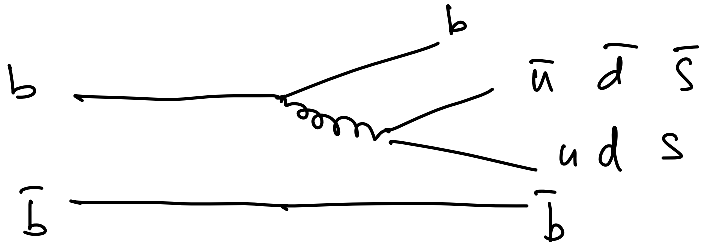
EM decay of $\Upsilon(1S)$.



Strong decay of $\Upsilon(1S)$.

$m_{\Upsilon(1S)} < 2m_B$.

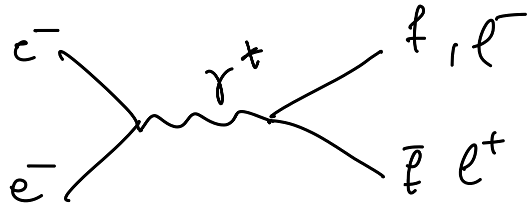
$\Upsilon(4S)$: $m = 10.58 \gtrsim 2m_B$



$\Upsilon(4S) \longrightarrow$ B^+B^- $b\bar{u}, \bar{b}u$
$B^0\bar{B}^0$ $b\bar{d}, \bar{b}d$
$\cancel{B_s^0 \bar{B}_s^0 \quad b\bar{s}, \bar{b}s}$

$$\boxed{2m_{B_s^0} > M_{\Upsilon(4S)}}$$

$\Upsilon(4S) \longrightarrow$ B^0B^0 50%
B^+B^- 50%



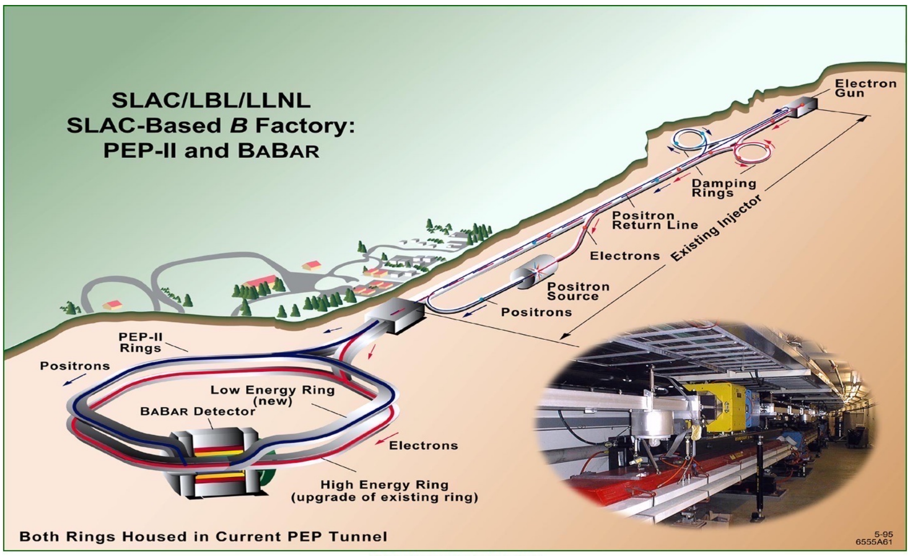
$\sqrt{s} = 10.58 \equiv M_{\Upsilon(4S)}$

$e^+e^- \longrightarrow b\bar{b} \equiv \Upsilon(4S)$
$c\bar{c}$
$s\bar{s}$
$u\bar{u}$
$d\bar{d}$
$\tau^+\tau^-$
$\mu^+\mu^-$
e^+e^-

$$\boxed{\begin{array}{c} \sigma(e^+e^- \to \Upsilon(4S) \to \text{tot}) \\ \simeq 3.5\ \text{nb.} \\ \\ \sigma(e^+e^- \to \Upsilon(4S) \to b\bar{b}) \\ \simeq 1.2\ \text{nb} \end{array}}$$

B-Factory:

	Collider	Exp.		$\mathrm{cm}^{-2}\mathrm{s}^{-1}$ Linst
SLAC, Stanford	PEP-II	BaBar	1999–2008	1.2×10^{34}
KEK, Japan.	KEK	Belle	1999 – 2010	2×10^{34}
	SuperKEK	Belle-II	202X –	5.2×10^{34}



PEP-II accelerator schematic and tunnel view

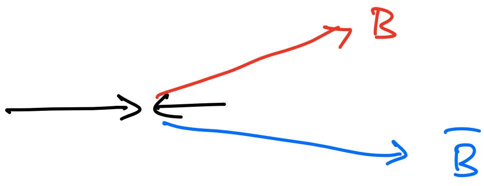
$\sqrt{s} = M_{\Upsilon(4S)}$
$= 10.58$ GeV

asymm.
e^+, e^- beams

$9.1\,\text{GeV}\ \xrightarrow{e^-}\quad \xleftarrow[3\,\text{GeV}]{e^+} \implies \beta\gamma = 0.56$

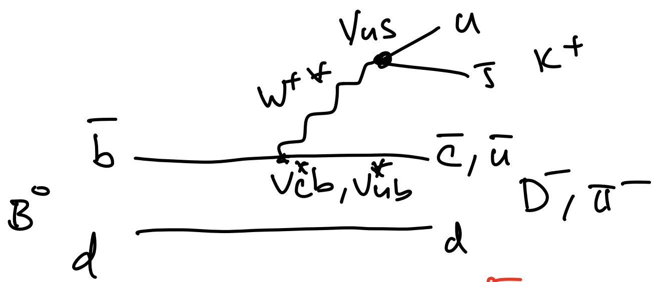
Lorentz boost of center-of-mass frame

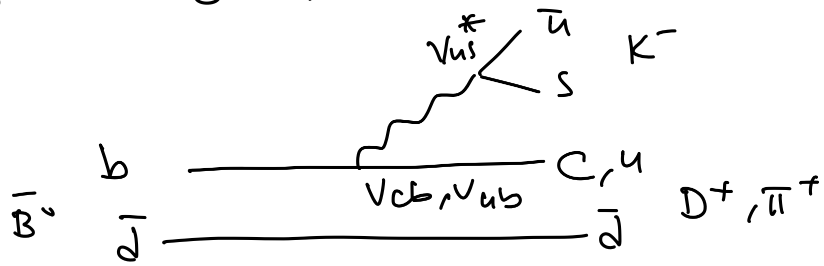
$e^+e^- \to \Upsilon(4S) \to B\bar{B}$



Direct CP violation

$B^0 \to D^- K^+ / \pi^- K^+ \overset{CP}{\implies} \bar{B}^0 \to D^+ K^- / \pi^+ K^-$

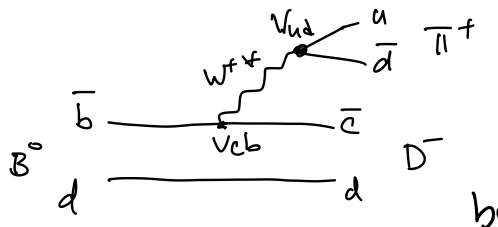




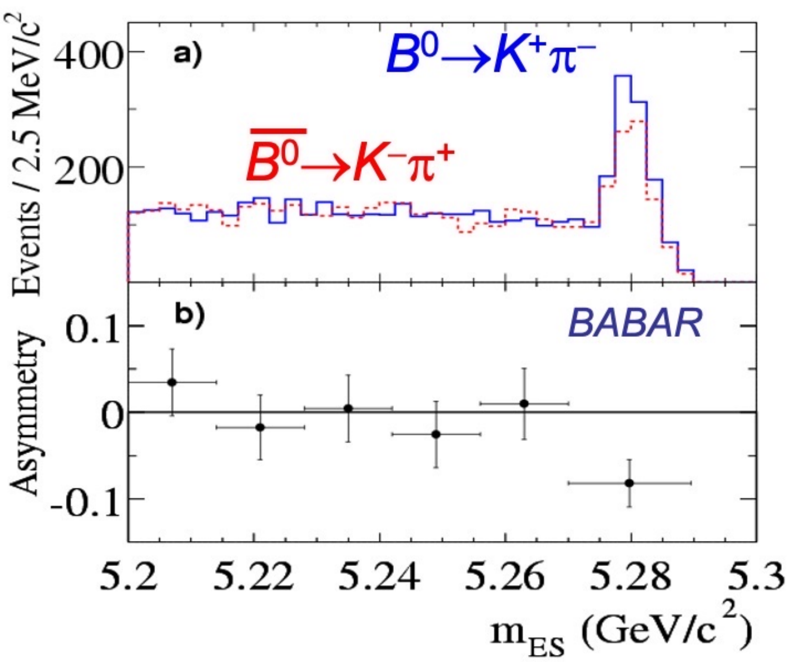
$BF(B \to K\pi) = 2\times10^{-5}$

$BF(B \to DK) = 2\times10^{-4}$

$N = \underset{\substack{\downarrow \\ 3.5 \\ \text{nb}}}{\sigma} \cdot \mathcal{L}_{\text{inst}} \cdot \Delta t \cdot \underset{\substack{\downarrow \\ B\bar{B}}}{25\%} \times \underbrace{\tfrac{1}{2}}_{B^0\bar{B}^0} \times BF(B \to DK) \times \underbrace{\Sigma\, \text{eff}_{\text{detector}}}_{\text{very high}}$



$BF(B \to D\pi) \simeq 2.5\times10^{-3}$
but no $\cancel{CP}$



original plot 2004

$n_{K\pi} = 1606 \pm 51$

$A_{K\pi} = -0.133 \pm 0.030 \pm 0.009$

$n(B^0 \to K^+\pi^-) = 910$

$n(\bar{B}^0 \to K^-\pi^+) = 696$

evidence for direct $\cancel{CP}$

oscillation of B^0 mesons

necessary for 2) $\cancel{CP}$ in mixing $B^0, \bar{B}^0$

3) interference

$$B^0 \longrightarrow f$$
$$B^0 \longrightarrow \bar{B}^0 \longrightarrow f$$

Oscillation of B mesons

$\Upsilon(4S)$

$$\bar{B}^0 \longleftarrow \bullet \longrightarrow B^0$$

$t=0 \qquad B^0, \bar{B}^0$

$\tau_B = 1.57 \; 10^{-15}\, s. \quad = 1.57 \; fs$

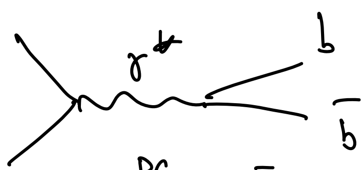
Suppose $\tau_B = \infty \Rightarrow$ no decay. only oscillation

$t=0 \qquad B^0, \bar{B}^0$

$|\psi, t=0\rangle = |B^0, \bar{B}^0\rangle$ initial state

$$|\psi, \text{final state}\rangle = |B_1 B_2\rangle = \frac{a|B^0\rangle|\bar{B}^0\rangle + b|\bar{B}^0\rangle|B^0\rangle}{\sqrt{a^2+b^2}}$$

$t=0$



$J^{PC} = 1^{--}$

$C = -1 \qquad C|B_1 B_2\rangle = -|B_1 B_2\rangle$

$\hookrightarrow \; b = -a$

$$|B_1 B_2\rangle = \frac{1}{\sqrt{2}}\left(|B^0\rangle|\bar{B}^0\rangle - |\bar{B}^0\rangle|B^0\rangle\right)$$ coherent/ Entangled State

B^0 flavor eigenstate $(\bar{b}, d) \neq$ weak/hamiltonian eigenstate.

if B^0 wel eigenstate $|B^0, t\rangle = |B^0\rangle e^{-iE_B t}$

in B^0 ref. frame: $|B^0, t\rangle = |B^0\rangle e^{-im_B t}$

if $\tau_B \neq 0 \Rightarrow |B^0\rangle = e^{-im_B t} e^{-\frac{\Gamma}{2} t} |B^0\rangle$.

$\Gamma = \frac{1}{\tau}$

Unfortunately $B^0, \bar{B}^0$ are not eigenstates of weak interactions.

B^0_L, B^0_H : weak eigenstates.

m_H, Γ_H heavy eigens.

m_L, Γ_L light eigenstate

assume $\Gamma_L = \Gamma_H = \frac{1}{\tau}$

$$|B^0\rangle = \frac{1}{\sqrt{2}}\left(|B_H\rangle + |B_L\rangle \right)$$
$$|\bar{B}^0\rangle = \frac{1}{\sqrt{2}}\left(|B_H\rangle - |B_L\rangle \right)$$
$$\Rightarrow$$
$$|B_L\rangle = \frac{1}{\sqrt{2}}\left(|B^0\rangle - |\bar{B}^0\rangle \right)$$
$$|B_H\rangle = \frac{1}{\sqrt{2}}\left(|B^0\rangle + |\bar{B}^0\rangle \right)$$

this notation assumes no $\cancel{CP}$ in mixing.

$$|B_L\rangle = p|B^0\rangle - q|\bar{B}^0\rangle$$
$$|B_H\rangle = p|B^0\rangle + q|\bar{B}^0\rangle$$

$p^2 + q^2 = 1$

most general notation allows for $\cancel{CP}$ in mixing

experimentally $\left|\frac{p}{q}\right| \sim 1$ @ 10^{-3} level.

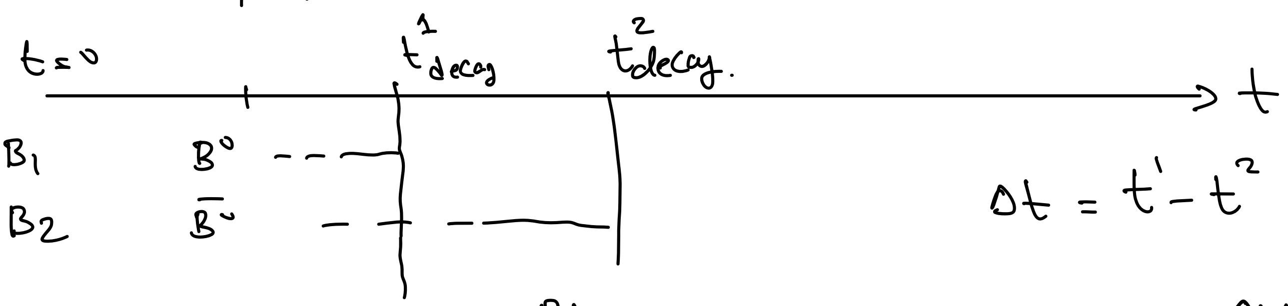
$\Rightarrow$ no $\cancel{CP}$ in mixing observed so far

$$|B_{L,H}(t)\rangle = e^{-im_{L,H} t} e^{-\Gamma_{L,H} t/2} |B_{L,H}\rangle$$

$\Rightarrow$ compute $|B^0, t\rangle$

$$|B^0, t\rangle = \frac{1}{\sqrt{2}}\left(|B_H, t\rangle + |B_L, t\rangle \right)$$

$$|B^0, t\rangle = \alpha(t)\,|B^0\rangle + \beta(t)\,|\bar{B}^0\rangle$$



$$\Delta t = t^1 - t^2 \gtrless 0$$

$$P(B^0 \to B^0) = \frac{e^{-\Gamma t}}{2}\left(1 + \cos(\Delta m\, \Delta t)\right)$$

$\Delta m =: m_H - m_L$

prob of no oscillation

$$P(B^0 \to \bar{B}^0) = \frac{e^{-\Gamma t}}{2}\left(1 - \cos(\Delta m\, \Delta t)\right)$$

prob of oscillation